

DANILA SHIRYAEV

Curriculum Vitae

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EDUCATION

PhD-Track, Advanced materials

Sep 2025 – Jun 2030

Institut Polytechnique, Palaiseau, France

Institut Polytechnique scholarship

Innovative Processes & Materials

GPA: 3.9/4.0

Bachelor of Arts, Chemistry

Aug 2023 – Dec 2024

Hunter College, City University of New York, NY, US

Hunter College Foundation scholarship

GPA: 3.9/4.0

Specialist, Chemistry (transferred to Hunter College)

Sep 2016 – Mar 2022

Moscow State University (MSU), Moscow, Russia

Full tuition merit-based scholarship with a monthly stipend

GPA: 3.6/4.0

RESEARCH EXPERIENCE

École Polytechnique

Sep 2025 – Mar 2026

Researcher, PMC laboratory, Delacroix's research group

Solid-State and Material Science research

- Synthesis and characterization of Eu-doped La-P-O phases mapped across the La_2O_3 - P_2O_5 phase diagram
- Investigation of inter-diffusion between LaPO_4 and EuPO_4 under laser treatment
- Transition rhabdophane/monazite LaPO_4 :Eu nanoparticles under laser

CUNY Research Foundation

Jan 2024 – Aug 2025

Research associate, Oliynyk's research group

Solid-State and Material Science research

- Work on intermetallics, synthesis and characterization of new compounds and their crystal structures
- Electronic structure calculations
- Using machine learning approaches to predict and systematize crystal structures of solids with classification
- Development of recommendation engine to target novel intermetallics materials

Moscow State University

Jan 2019 – Mar 2022

Research assistant, Badun's research group

Radionuclides and Labeled Compounds research

- Preparation of tritium-labeled organic compounds using the thermal activation method
- Investigation by radionuclide methods of adsorption phenomena at liquid/liquid interface

Moscow State University

Sep 2016 – Dec 2019

Research assistant, Pankratov's research group

Nuclear Chemical Materials Science research

- Synthesis of compounds of Fe in unstable oxidation states with given functional properties
- Development of materials and methods for reagent and reagent-free water treatment
- Application of absorption and emission Mössbauer spectroscopy to solid state research

PUBLICATIONS

Shiryaev, D.; Jaffal, E. I.; Lee, S.; Selvaratnam, B.; Oliynyk, A. O. Achieving a Scalable Machine Learning Workflow for Crystal Structure Discovery with Experimental Validation. *Digital Discovery* 2026, 5 (6), 2414–2437. DOI: [10.1039/D6DD00132G](https://doi.org/10.1039/D6DD00132G)

Pozdnyakova, N.; Jaffal, E. I.; **Shiryaev, D.;** Xhabrahimi, B.; Selvaratnam, B.; Oliynyk, A. O. RE₇M₄InGe₁₂ (RE = Y, Gd–Er; M = Rh, Os) Quaternary Germanide Series Expansion with Structural Motif Clustering. *Inorg. Chem.* 2026, 65 (1), 517–531. DOI: [10.1021/acs.inorgchem.5c04639](https://doi.org/10.1021/acs.inorgchem.5c04639)

Jaffal, E. I.; **Shiryaev, D.;** Selvaratnam, B.; Oliynyk, A. O. Investigating Mechanical Properties through Defect Chemistry in Hard Binary Phosphide Material Ta₃P. *Solid State Commun.* 2026, 408, 116250. DOI: [10.1016/j.ssc.2025.116250](https://doi.org/10.1016/j.ssc.2025.116250)

Xhabrahimi, B.; Jaffal, E. I.; **Shiryaev, D.;** Barua, N. K.; Donohoe, M.; Pozdnyakova, N.; Ismail, M.; Selvaratnam, B.; Niknam, E.; Kleinke, H.; Oliynyk, A. O. Explainable Recommendation Engines to Predict Complex Intermetallics: Synthesis and Characterization of Gd₁₀RuCd₃, a Neutron Absorption Material. *J. Am. Chem. Soc.* 2025, 147, 36589–36603. DOI: [10.1021/jacs.5c11646](https://doi.org/10.1021/jacs.5c11646)

Selvaratnam, B.; Jaffal, E. I.; **Shiryaev, D.;** Oliynyk, A. O. Dataset of Prototype Structures Adopted by Intermetallic Compounds with AB Stacking. *Data in Brief* 2025, 112138. DOI: [10.1016/j.dib.2025.112138](https://doi.org/10.1016/j.dib.2025.112138)

Shiryaev, D.; Selvaratnam, B.; Sun, Y.; Jaffal, E. I.; Oliynyk, A. O. Exploring Feature Engineering for Crystal Structure Classification: Interactive Applications of PCA and PLS-DA Clustering. *J. Chem. Educ.* 2026, 103 (1), 662–670. DOI: [10.1021/acs.jchemed.5c00723](https://doi.org/10.1021/acs.jchemed.5c00723)

Sethi, S. S.; Dutta, A.; Jaffal, E. I.; Yadav, N.; **Shiryaev, D.;** Hoang, B.; Machathi, A.; Lee, S.; Das, K.; Jana, P. P.; Oliynyk, A. O. Unsupervised Machine Learning Prediction of a Novel 1:3 Intermetallic Phase with the Synthesis of TbIr₃ (PuNi₃-Type) as Experimental Validation. *J. Am. Chem. Soc.* 2025, 147 (17), 14739–14755. DOI: [10.1021/jacs.5c03510](https://doi.org/10.1021/jacs.5c03510)

Jaffal, E. I.; Lee, S.; **Shiryaev, D.;** Vtorov, A.; Barua, N. K.; Kleinke, H.; Oliynyk, A. O. Composition and Structure Analyzer/Featurizer for Explainable Machine-Learning Models to Predict Solid State Structures. *Digital Discovery* 2025, 4 (2), 548–560. DOI: [10.1039/D4DD00332B](https://doi.org/10.1039/D4DD00332B)

RELATED WORK EXPERIENCE

Saint-Gobain Innovation Hub, Paris, France

Mar 2026 – Aug 2026

Research Intern

- Optimized data acquisition and developed a protocol for photoluminescence mapping on a confocal Raman setup coupled with a cryostat for reliable temperature-dependent measurements
- Performed measurements and modeling to estimate excitation laser penetration depth to support data interpretation
- Built Python pipelines to automate processing, decomposition, and analysis of large multispectral datasets, accelerating data turnaround across the project

Research Foundation CUNY, New York, US

Oct 2024 – Aug 2025

Research Associate

- Mechanical property optimization through defect chemistry of pnictides. Research in enhancing the hardness, wear, and corrosion resistance of tantalum-based materials for oil drilling applications.
- Focus on defect engineering through pnictogen doping to improve durability and performance under extreme conditions, combining experimental synthesis with X-ray diffraction, DFT calculations, and mechanical testing
- Mentoring other undergraduate students in their laboratory investigation

Evotech-Mirai Genomics, Moscow, Russia

Jul 2021 – Dec 2021

Clinical application manager

- Trained over 100 new customers on device functionality, resulting in a 15% increase in customer satisfaction ratings
- Successfully launched and managed over 20 new laboratories in Russia and Kazakhstan, expanding the market presence by 20%

EuroChem, Nevinnomyssk, Russia

Jun 2021 – Jul 2021

Absorption operator

- Operated absorption unit, ensuring smooth and efficient operation of critical processes, hourly sampling
- Detailed study of all processes occurring at each unit on NPK fertilizers production

“Archimedes”, Moscow, Russia

Oct 2020 – May 2021

PCR laboratory assistant

- Responsibilities included the complete testing cycle, from processing samples to analyzing the results
- Conducted a major optimization of work in the laboratory, thereby improving and structuring workflow, reporting, improving conditions for both customers and staff

PRESENTATIONS**ACS Mid-Atlantic Regional Meeting – South Orange, NJ**

May 2025

“Visualization and recommendation framework for targeting novel solid-state materials”

POSTERS**Hunter Research Conference – New York, NY**

Apr 2025

“Structure type explorer (STEx): visualization and recommendation approach to target novel solid-state materials”

HACKATHONS**SSMC-Collaboration Incubator – Madison, WI**

May 2025

Selected participant for national hackathon-style research workshop. Collaborated on the *Rational Design of Thermoelectrics, Light Emitting Materials* with an interdisciplinary cohort of PhD students and professors.**TEACHING EXPERIENCE****Moscow State University**

Fall 2021

Teaching Assistant

Radiochemistry lab course with Dr. Badun

STUDENTS MENTORED**Sviatoslav Pisarev** (B.A. Chemistry, 2024)**Brook Xhabrahimi** (B.A. Chemistry, 2025)**Natalia Poznyakova** (B.A. Chemistry, 2025)**Miriam Ismail** (B.A. Chemistry, 2025)**Yujing Sun** (The Bronx High School of Science, 2025)**GRANTS AND SCHOLARSHIPS**

Saint-Gobain & École Polytechnique Chair of Innovative Processes & Materials (€5000)

2025

Institut Polytechnique scholarship (€10000)

2025

Scholarship from Hunter College Foundation (\$8300)

2024

Scholarship from Hunter College Foundation (\$13500)

2023

TECHNICAL SKILLS

Software: Bruker Suite, Topas, Match, VESTA, Diamond, Pearson's Crystal Data, TopSpin, VASP, Microsoft Office.

Programming & markup languages: Python, Bash, HTML/CSS, Markdown.

Languages: Russian (native), English (full professional proficiency), French (elementary)

MEMBERSHIPS

American Chemical Society (ACS)	2024 - present
Material Research Society (MRS)	2024 - present